

EBN 122 - Schedule 2023

Abbreviations: PdV - Prof Pieter de Villiers; PJ - Prof Pieter Jacobs; ALs - Assistant Lecturers; A01 – Assignment 1, etc.

Lectures: **Mo** 12:30-13:30 (AE du Toit Auditorium); **Tu** 07:30-08:20 (AE du Toit Auditorium); **Fr** 07:30-8:20 (AE du Toit Auditorium)

Tut classes: **We** 09:30-10:20 Eng III-6, Eng III-7

Week	Day	Lecture/Tut Class/Class Test	Sections in Textbook / End-of-Chapter Problems		Remarks
1	Mo	24-Jul	Study Guide & Lecture 1	1.1 – 1.4	Units, Charge and Current, Voltage
	Tu	25-Jul	Lecture 2 <i>A01 released</i>	1.5 – 1.6	Power and Energy, Circuit Elements
	We	26-Jul	Tut Class # 1	1.2 – 1.6	1.4, 1.6, 1.8, 1.11, 1.12, 1.16, 1.18, 1.20, 1.24, 1.28 OCR answer sheet
	Th	27-Jul			
	Fr	28-Jul	Lecture 3	2.2 – 2.3	Ohm's Law, Nodes & Branches & Loops
2	Mo	31-Jul	Lecture 4	2.4	Kirchoff's Law
	Tu	1-Aug	Lecture 5 <i>A01 due</i>	2.5 – 2.6	Series and Voltage Division, Parallel and Current Division
	We	2-Aug	Tut Class # 2	2.2-2.6	2.6, 2.7, 2.15, 2.22, 2.23, 2.25, 2.32, 2.45b
	Th	3-Aug			Pre-prac lecture for Practical 1
	Fr	4-Aug	Lecture 6 <i>A02 released</i>	2.7	Wye-Delta transformations
3	Mo	7-Aug	Lecture 7	3.2	Nodal Analysis
	Tu	8-Aug	Lecture 8	3.3	Nodal Analysis with Voltage Sources
	We	9-Aug	Public Holiday		
	Th	10-Aug			
	Fr	11-Aug	Tut Class # 3 <i>A02 due</i>	2.7, 3.2-3.3	2.51b, 2.53b, 3.2, 3.4, 3.12, 3.16, 3.20, 3.30, 3.31
4 Practical 1 Week	Mo	14-Aug	Lecture 9	3.4	Mesh Analysis
	Tu	15-Aug	Lecture 10	3.5	Mesh Analysis with Current Sources
	We	16-Aug	Class Test #1		
	Th	17-Aug			
	Fr	18-Aug	Lecture 11 <i>A03 released</i>	3.6 – 3.7	Nodal and Mesh Analysis by Inspection
5	Mo	21-Aug	Lecture 12	3.9	Transistors
	Tu	22-Aug	Lecture 13 <i>A04 released</i>	4.2 – 4.3	Linearity and Superposition
	We	23-Aug	Tut Class # 4	3.4-3.7, 3.9	3.37, 3.44, 3.60, 3.69, 3.72, 3.90, 3.91
	Th	24-Aug			
	Fr	25-Aug	Lecture 14 <i>A03 due</i>	4.4	Source Transformations
6	Sa 26 Aug— Sa 2 Sep		Test Week 1		
7	Mo	4-Sep	Lecture 15	4.5	Thevenin's Theorem
	Tu	5-Sep	ST1 Discussion <i>A04 due</i>		
	We	6-Sep	Tut Class # 5	4.2-4.4	4.4, 4.5, 4.11, 4.18, 4.24
	Th	7-Sep			Pre-prac lecture for Practical 2
	Fr	8-Sep	Lecture 16	4.6	Norton's Theorem
8	Mo	11-Sep	Lecture 17	4.8	Maximum Power Transfer (Wednesday timetable)
	Tu	12-Sep	Lecture 18	5.2 – 5.3	Op Amps, Ideal Op Amps (Monday timetable)
	We	13-Sep	Spring Day		
	Th 14—Su 24 Sep		Recess		
9	Mo	25-Sep	Heritage Day		
	Tu	26-Sep	Lecture 19 <i>A05 released</i>	5.4 – 5.5	Inverting and Non-inverting Amplifiers
	We	27-Sep	Tut Class # 6	4.5-4.8 & 5.2-5.3	4.37, 4.38, 4.40, 4.48, 4.52, 4.71, 4.72, 5.6, 5.8, 5.9
	Th	28-Sep			
	Fr	29-Sep	Lecture 20	5.6 – 5.7	Summing and Difference Amplifiers

10 Practical 2 Week	Mo	2-Oct	Lecture 21	5.8	Cascade Circuits	PJ
	Tu	3-Oct	Lecture 22 A05 due	6.2 – 6.3	Capacitors, Series and Parallel Capacitors	PJ
	We	4-Oct	Tut Class # 7	5.4-5.8	5.13, 5.20, 5.31, 5.32, 5.39, 5.45, 5.57, 5.70	ALs
	Th	5-Oct			Pre-prac lecture for Practical 3	VL
	Fr	6-Oct	Class Test #2			
11	Mo	9-Oct	Lecture 23 A06 released	6.4 – 6.5	Inductors, Series and Parallel Inductors	PJ
	Tu	10-Oct	Lecture 24	App B	Complex Numbers	PJ
	We	11-Oct	Tut Class # 8	6.2—6.5 & App B	6.10, 6.12, 6.18,6.40,6.46, 6.54	ALs
	Th	12-Oct				
	Fr	13-Oct	Lecture 25 A07 released	9.2 – 9.3	Sinusoids & Phasors	PJ
12	Mo 16—Sa 21 Oct		Test Week 2			
13 Practical 3 Week	Mo	23-Oct	Lecture 26 A06 due	9.3 – 9.4	Phasors, Phasor Relationships with Circuit Elements	PJ
	Tu	24-Oct	Lecture 27 A08 released	9.5 – 9.7	Impedance, Kirchoff's Laws in Frequency Domain	PJ
	We	25-Oct	Tut Class # 9	9.2 – 9.7	9.2, 9.6, 9.13, 9.19, 9.20, 9.42, 9.48, 9.54, 9.66, 9.70	ALs
	Th	26-Oct				
	Fr	27-Oct	ST2 Discussion A07 due			
14	Mo	30-Oct	Lecture 28	10.2	Nodal Analysis	PJ
	Tu	31-Oct	Lecture 29 A09 released	10.3 – 10.4	Mesh Analysis, Superposition	PJ
	We	01-Nov	Class Test #3			
	Th	02-Nov	Lecture 30	10.5 – 10.6	Source Transformation, Thevenin & Norton	PJ
	Fr	03-Nov	Tut Class # 10 A08 due	10.2 –10.6	10.15, 10.19, 10.36, 10.46, 10.52, 10.56, 10.61, 10.66, 10.68 (In AE du Toit Auditorium)	ALs
15	Mo	06-Nov	Revision A09 due		Semester Test 3 (optional)	PJ
	Tu	07-Nov				
	We	08-Nov				
	Th	09-Nov				
	Fr	10-Nov				

Assignment topics

A01 Chapter 1

A02 Chapter 2

A03 Chapter 3, 3.2 – 3.5

A04 Chapter 3, 3.6, 3.7, 3.9

A05 Chapter 4, 5.2 – 5.5

A06 Chapter 5.6 – 5.10, 6

A07 Chapter 9.2 – 9.3

A08 Chapter 9.4 – 9.8

A09 Chapter 10.2 – 10.7